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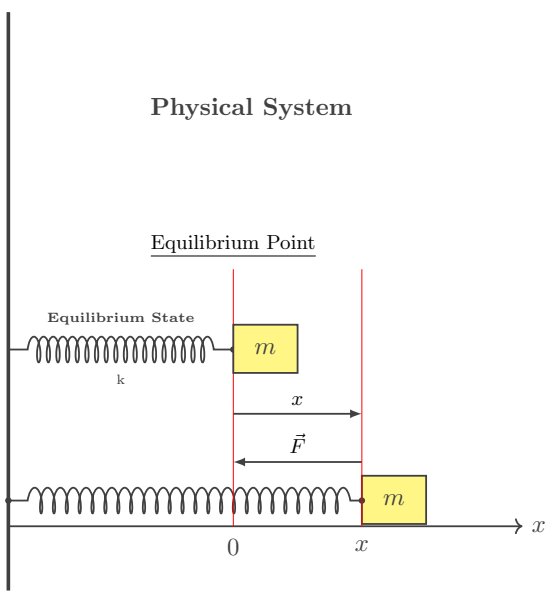


Hooke's Law

$$\mathbf{F} \propto -\mathbf{x}$$

Hooke's Law with Spring Constant k

$$\mathbf{F} = -\mathbf{kx}$$

Newton's 2nd Law of Motion

$$\mathbf{F} = \mathbf{ma}$$

Newton's Force Equals Hooke's Force

$$\mathbf{ma} = -\mathbf{kx} \Rightarrow \mathbf{a} = -\frac{\mathbf{k}}{\mathbf{m}}\mathbf{x}$$

Equation of Motion

$$\boxed{\mathbf{a} = \ddot{\mathbf{x}} = \frac{\mathbf{d^2x}}{\mathbf{dt^2}} = -\frac{\mathbf{k}}{\mathbf{m}}\mathbf{x}}$$

Figure 1: Simple Spring and Mass System with Spring Constant k and Mass m

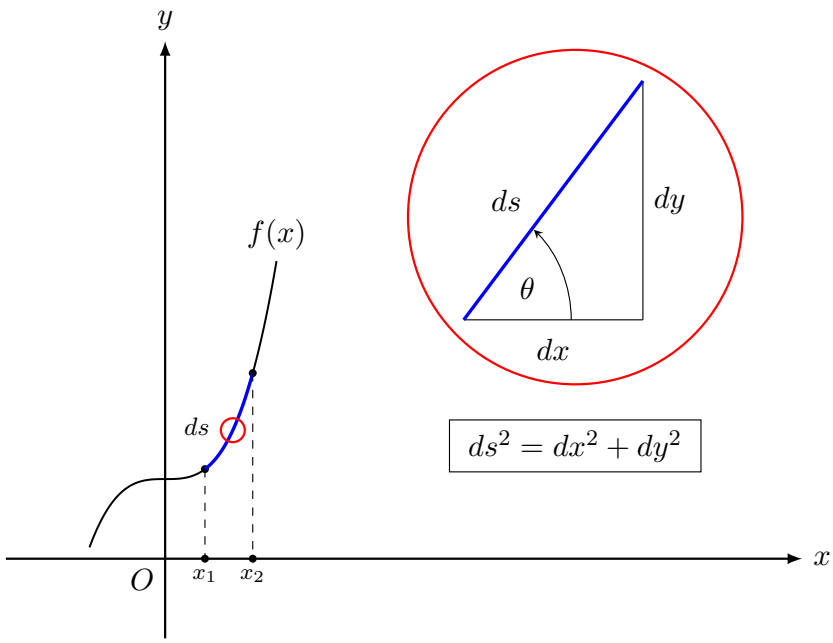


Figure 2: $f(x)$, ds and the Pythagorean Theorem

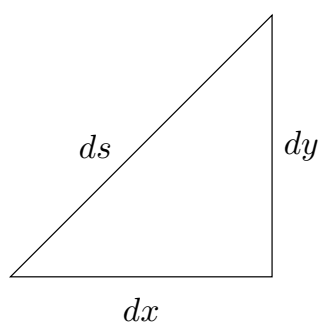


Figure 3: $ds^2 = dx^2 + dy^2$

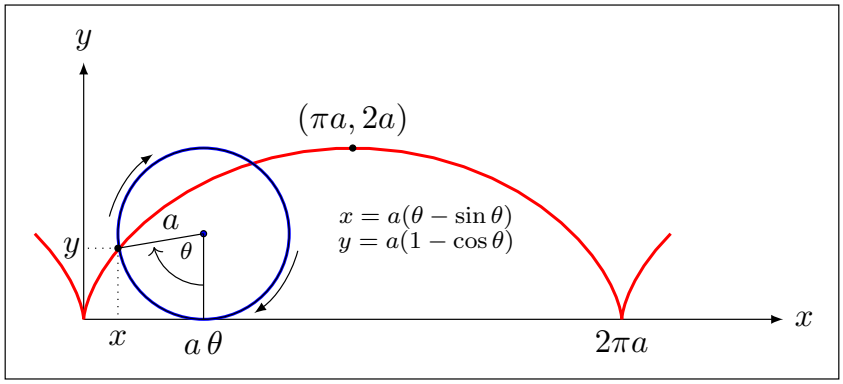


Figure 4: The Cycloid Curve

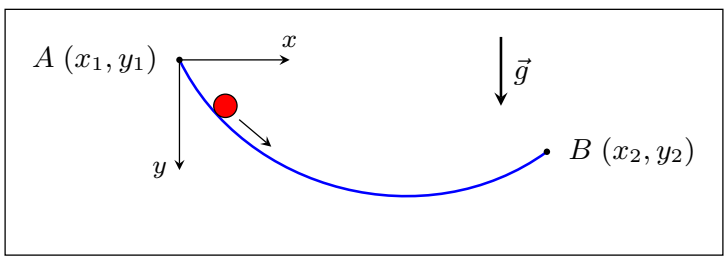


Figure 5: The Brachistochrone Problem Setup

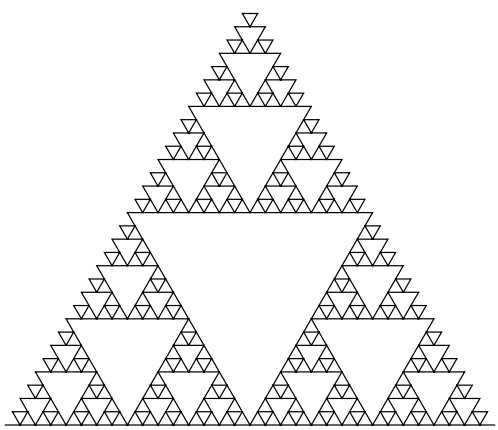


Figure 6: Sierpiński triangle/gasket

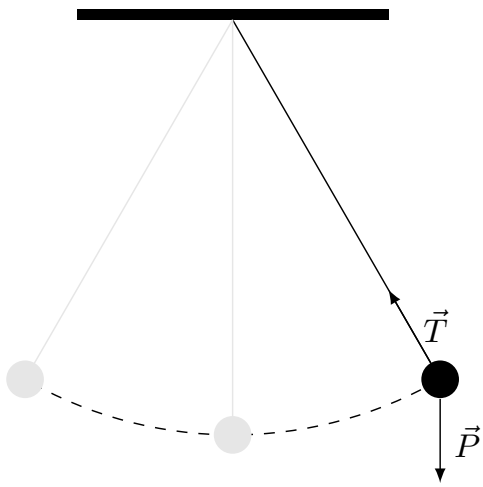


Figure 7: Pendulum

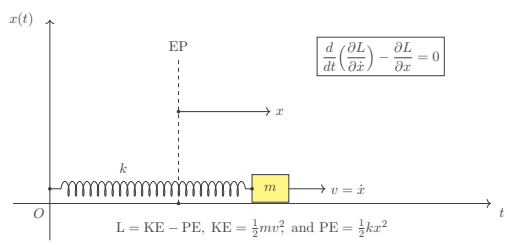


Figure 8: Simple Harmonic Motion Setup

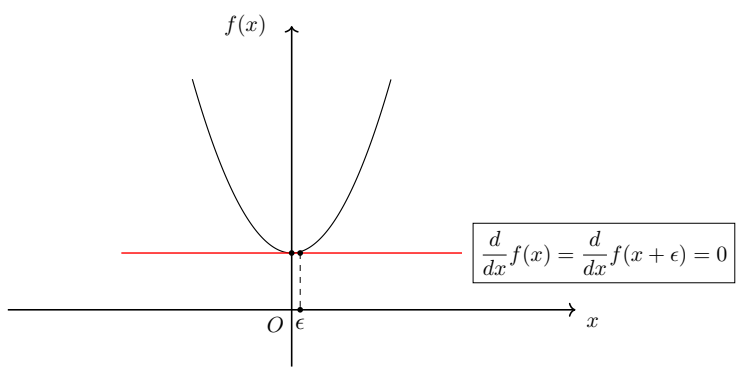


Figure 9: $f(x) = x^2 + 1$

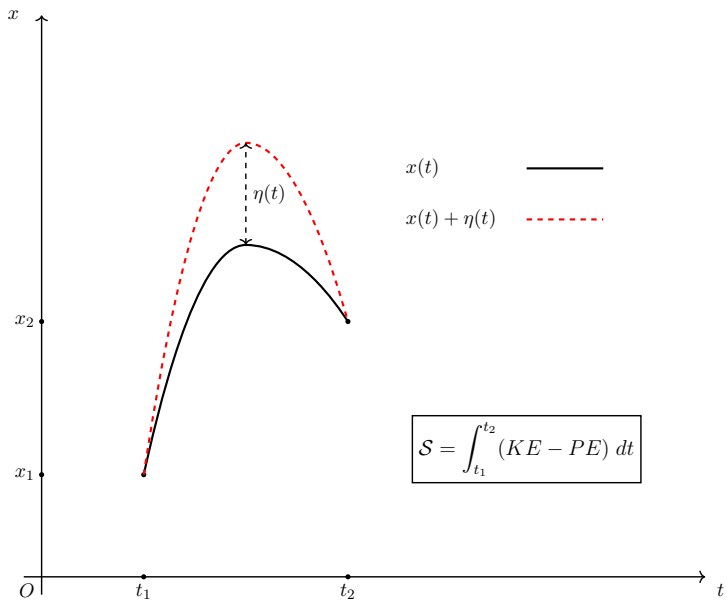


Figure 10: Principle of Least Action Setup

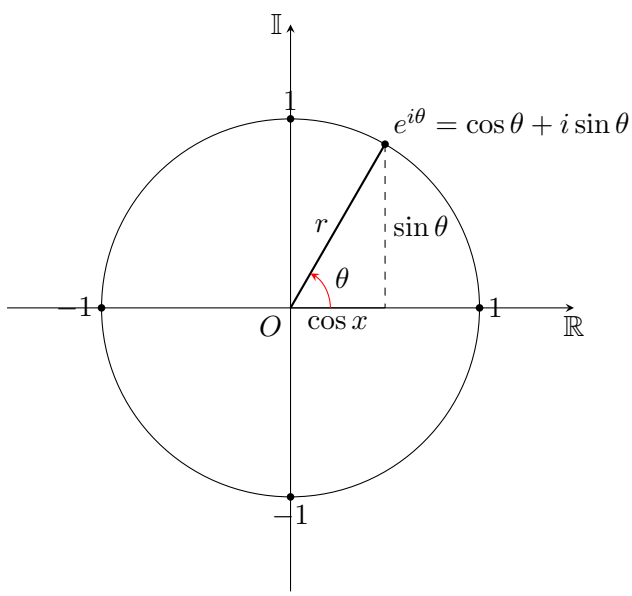


Figure 11: Euler's Formula in the Complex Plane

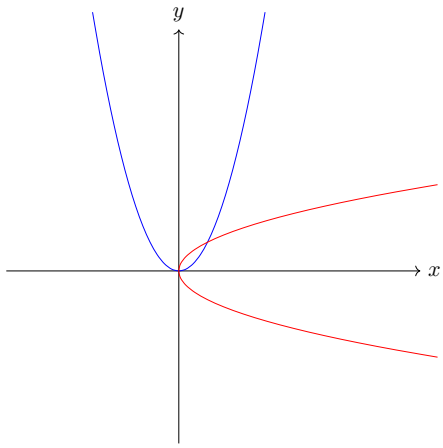
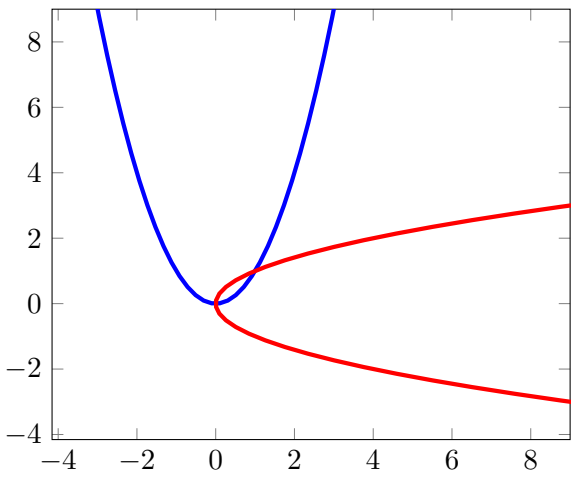
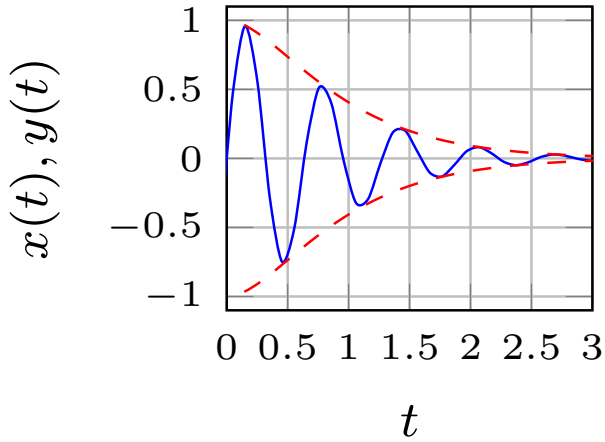
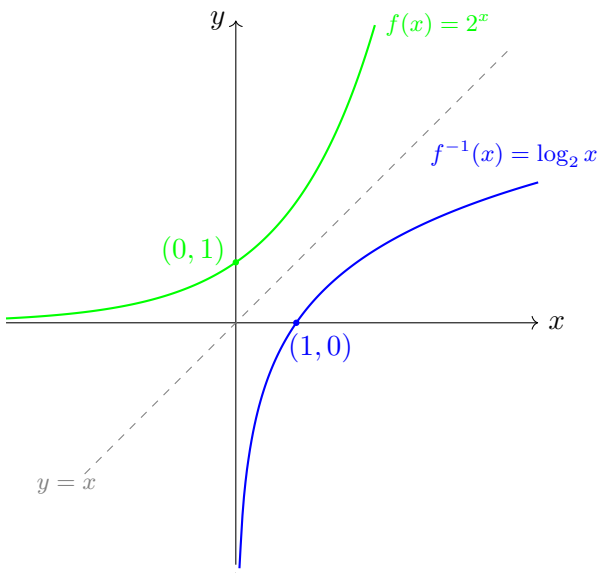
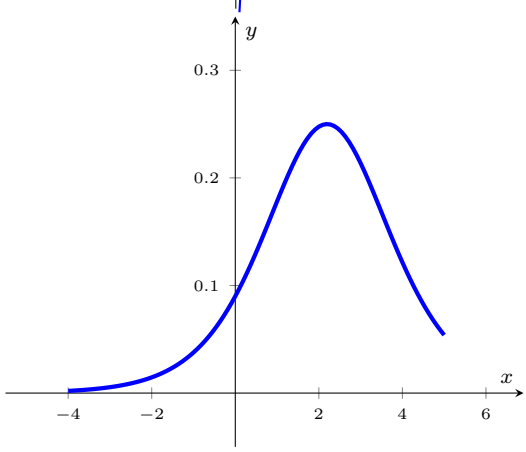


Figure 12: This is a graphing test




1
0.5
0
−0.5
−1
0 0.5 1 1.5 2 2.5 3
$x(t), y(t)$
t


y
$f(x) = 2^x$
$f^{-1}(x) = \log_2 x$
$(0, 1)$
x
$(1, 0)$
$y = x$


y
0.3
0.2
0.1
x
−4
−2
2
4
6